A Cyber Mission Impact Assessment Tool

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***Abstract* – The promise of practicing mission assurance is to be able to leverage an understanding of how mission objectives and outcomes are dependent on supporting cyber resources. This makes it possible to analyze, monitor, and manage your cyber resources in a mission context. In previous work, we demonstrated how process modeling tools can simulate mission systems to allow us to dynamically compute the mission impacts of cyber events. We demonstrated the value of using this approach, but unfortunately practical deployment of our work was hampered by limitations of existing commercial off-the-shelf (COTS) tools for process modeling. To address this deficiency, we have developed our own Cyber Mission Impact Business Process Modeling tool. Although it implements only a functional subset of the business process modeling notation (BPMN), it has, unlike the more generic COTS tools, been specifically designed for the representation of cyber processes, resources, and cyber incident effects. The method and tool are described in this paper.**

Keywords – mission assurance, mission impact, cyber effects, impact assessment, simulation, business process modeling

I. INTRODUCTION

The promise of practicing mission assurance is to be able to leverage an understanding of how mission objectives and outcomes are dependent on supporting cyber resources [1]. This makes it possible to analyze, monitor, and manage ones cyber resources in a mission context rather than just tend them as jobs unto themselves. The cyber mission impact assessment (CMIA) approach and tool we describe helps realize this vision, making it possible to model missions in a way that estimates the impact of cyber incidents.

CMIA is a foundational aspect of any cyber mission risk assessment. We have described previously how a systems' cyber resilience can be equated with its inverse, cyber mission risk [8]. From a systems engineering perspective, CMIA makes it possible to perform system assessments by simulating the application of potential security and resilience methods to a system within the mission context. Since effective resilience methods either prevent or mitigate the impacts of cyber incidents, when combined with the probability that bad events will occur, the impacts computed by CMIA address the "amount of loss" part of the risk equation. In a forthcoming work we will describe how our CMIA tool can be combined with a topological attack model to support mission assurance assessments and return-on-investment calculations. In this paper, however, we focus on the CMIA tool itself, which already provides value in its own right.

The key to accomplishing cyber mission impact assessment is to represent the mission and its cyber dependencies. In previous work [2][3], we showed how the simulation capability of process modeling tools can allow one to dynamically compute the mission impacts of cyber events. We demonstrated the value this approach over more static representations of mission systems. Unfortunately, practical deployment of our approach was hampered by limitations of existing COTS tools that support process modeling. Available products, such as iGrafx [4] and ExtendSim [5], have the necessary capabilities to demonstrate our approach but not to support more real-world employment of it. To address this deficiency, we have developed our own Cyber Mission Impact Business Process Modeling tool. Although it implements only a functional subset of BPMN, it has, unlike the more generic COTS tools, been specifically designed for the representation of cyber processes, resources and cyber incident effects. As such, it more naturally supports the functionality needed for cyber mission impact assessments and makes it unnecessary for modelers to clutter a model with extraneous content that ends up making those models harder to develop, comprehend, and maintain once they have been built.

The paper is organized as follows. The next section describes how business process models support creating mission-level models (MLM) and how the mission systems IT dependencies can be captured. It is followed by a section that describes how business process modeling can be used to represent and simulate cyber incident impacts. In our software these incident representations are now automatically created, whereas they must be manually added in COTS tools. The next section discusses our custom software, and the paper ends with a discussion of our work.

II. MISSION-LEVEL MODELING

Business process models can be used to represent mission systems in the context of their execution of mission threads. A mission thread represents a precise, objective description of a task. In other words, a mission thread is a time-ordered, operational event description that captures discrete, definable interactions among mission resources, such as human operators and technological components. Progress has been made in executable standard modeling languages, such as BPMN [6], which are suitable for capturing mission threads. The BPMN standard contrasts with historical trends of proprietary languages in simulation tools such as OPNET and EXTEND. An advantage of a language such as BPMN is that is allows non-engineers, who are primarily interested in assessing mission outcomes, to model and simulate system behavior using an intuitive, graphical modeling language. Its graphical nature makes it easy for SMEs to understand and confirm the model's adequacy in a way that would not be possible by showing them computer code. After defining testable measures of effectiveness (MOE), measures of

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performance (MOP), and key performance parameters (KPP) for the modeled mission, the process model captures how the performance of mission activities contributes to achieving them. It then becomes possible to use a BPMN model as a simulation to analyze a mission system.

and money, and it is these types of improvements that a business analyst will look for.

An example of the type of capability provided by these executable process models is shown in Figure 1. It shows event generators that can generate mission instances that traverse the

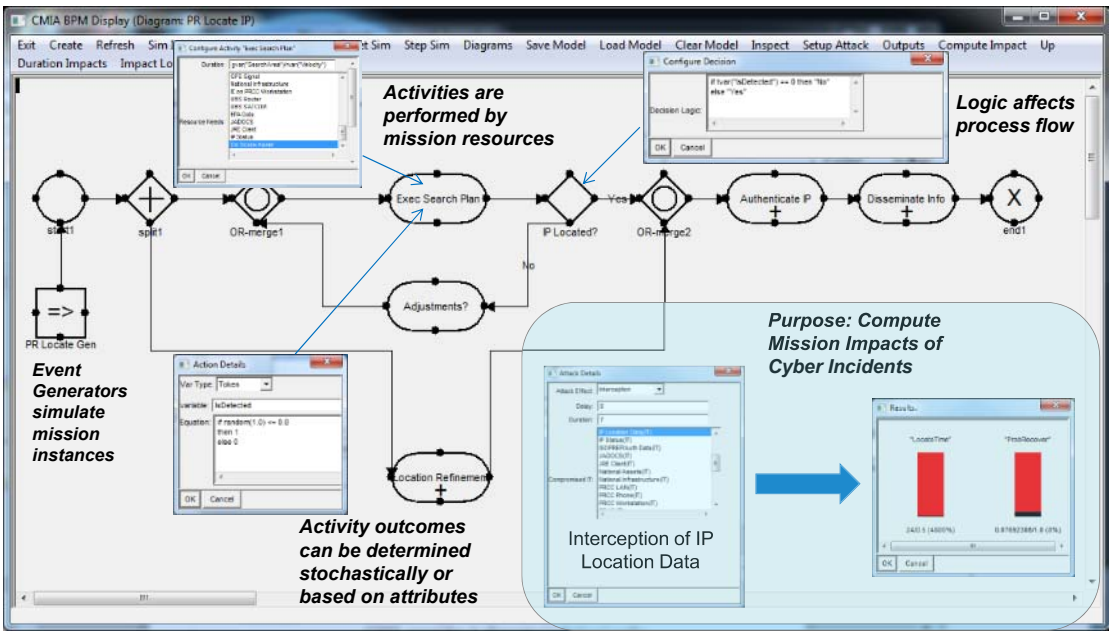

Figure 1: MITRE's CMIA Software Supports Mission-Level Modeling and Cyber Mission Impact Assessment

Mission threads can be modeled as a MLM using BPMN and can be simulated and analyzed using COTS software such as iGrafx. This allows one to evaluate mission systems using modeling and simulation early in the design process and acquisition cycle. These modeling technologies allow one to

process work-flow as transaction-style tokens. Resources that represent human, information and communications technologies (ICT), or other components are assigned to perform activities. Attributes can be assigned to global variables, to the transaction tokens, and to resources. Activity timeframes and outcomes can

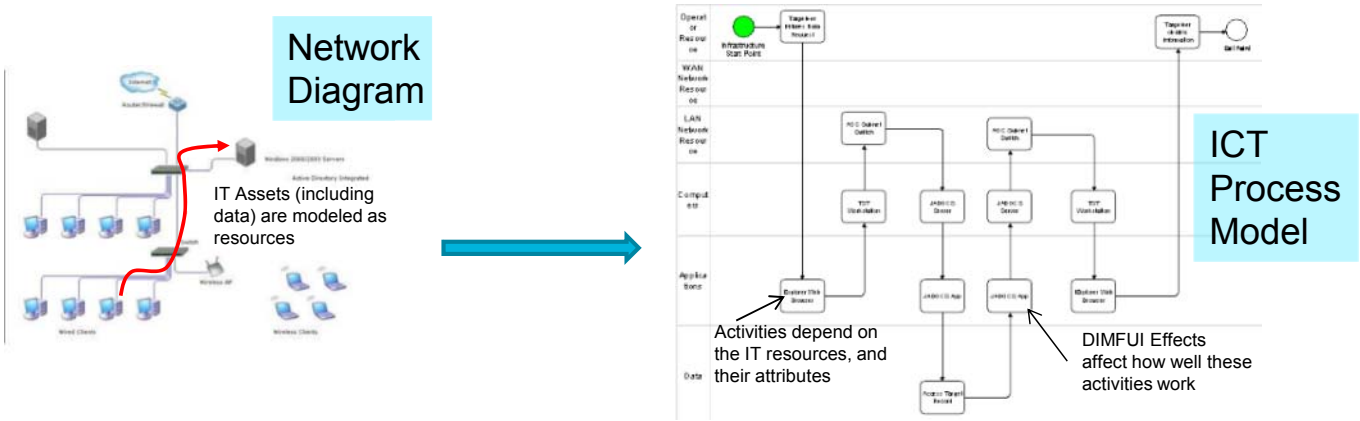

Figure 2: A Process Model of an ICT Activity

assess and/or improve a system given the selected mission context. Since a process model captures activity, control, and information flows, it is possible to evaluate alternate variations on resource assignments, information, and control flows and to assess potential improvements. Using a faster resource can speed up a system and increase capacity. A decision to route fewer mission instances through a costly approval chain can save time

be determined by computing with attribute values or stochastically. Logic statements determine process flow through the model.

CMIA models must include mission systems dependencies on ICT. As is depicted in Figure 2, ICT resources perform tasks just like any other mission resource. In the diagram's left is a

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user using an application running on a client host, which connects to a switch, which connects to another switch, which connects to a server, which runs a service that accesses some data that is needed by the client. The right in Figure 2 shows the action sequence the ICT performs. To succeed, all of the ICT resources involved must be functioning and performing their individual tasks. Unlike tree-structured representations of these ICT dependencies, the process model representation accurately captures multiple (redundant) communications paths, as well as fallback and failover processes that automatically kick in when a failure or incident occurs.

A CMIA model must characterize the ability of these ICT activities to be performed in the face of cyber incidents. Rather than consider all of the possible attack instances, as is described in [7], we instead consider only the effects created by successful cyber-attacks creating one of six effects: Degradation, Interruption, Modification, Fabrication, Unauthorized Use, and Interception (DIMFUI). These represent a comprehensive set of cyber effects that could be caused by a cyber incident. Interception, modification, and interruption are commensurate with the confidentiality, integrity and availability effects that many use. Degradation is the ability to reduce performance, capacity, or quality. Fabrication is the ability to inject false signals, data, or components. Unauthorized-use is the ability to gain unauthorized access or perform non-approved actions [7].

A CMIA model turns a cyber incident effect into a mission effect. This considers that the incident has occurred and has resulted in one or more DIMFUI effects against one or more of the cyber resources. Contrary to the statements by Barreto [13], CMIA models easily accommodate estimating multiple simultaneous incident effects against multiple cyber resources. CMIA models should accommodate any incident effects that can occur, irrespective of whether or not there are sensors that can actually detect them. In this way, the models represent a ground truth that can allow them to be used to assess the mission system. This is further described in [2][3][1].

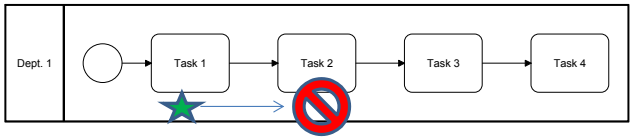

Figure 3: Cases 1 & 2 - A mission thread needs use of an attack-affected resource

III. BPMN Representation of Cyber Incidents

Process models can be used to inform reasoning about mission outcomes in the face of cyber incidents. Here, we describe how this is done in BPMN, even though our CMIA software implements these attack processes for you automatically. A number of process-oriented use cases determine whether or not the effect of a cyber incident can cause mission impacts. These are described below:

Figure 3 shows a sequence of mission tasks. Mission impact Case 1 shows that a mission thread is executing Task 1 and that the ICT resource needed for Task 2 has been affected by a cyber incident. In this case impact may occur since Task 2 is performed before the affected resource has recovered from the incident. In Case 2 no impact occurs as a result of the incident because the incident ends, or is recovered from before Task 2 is performed by the mission thread. So the timing and duration of incidents on mission thread instances is important.

In Figure 4, the mission thread is performing Task 3, and the

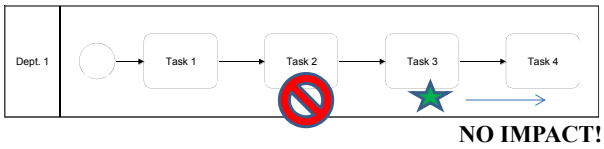

Figure 4: Case 3 – The mission thread no longer needs the affected resource

ICT resource required for Task 2 is affected by an incident. In this situation no impact occurs since the mission thread no longer needs the affected resource to complete.

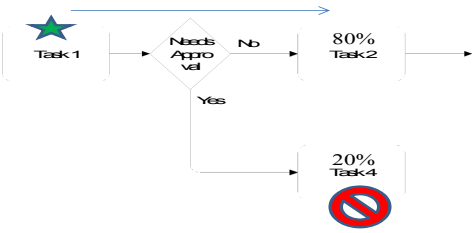

Figure 5: Case 4 & 5 – Whether a mission thread follows the process path affected by the attacked

In Figure 5, whether or not impact can occur will depend on the workflow path the mission thread takes. In Case 4 no impact occurs because the mission thread takes the workflow path that does not require the affected ICT resource. In Case 5 impact can occur because the mission thread takes the workflow path that requires the affected ICT resource.

In Figure 6, Case 6 shows that the use of parallel workflow paths joined by an inclusive OR gate should not cause impact so long as there is a workflow path that reaches the OR that doesn't make use of the ICT resource affected by the attack.

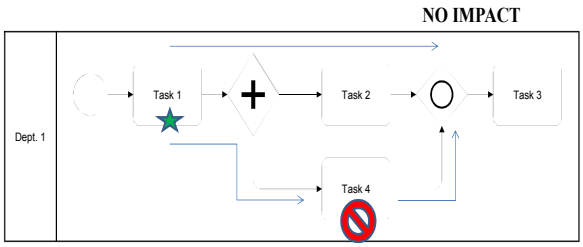

Figure 6: Case 6 – Inclusive OR allows the mission thread to proceed if any one of the possible paths succeed

In Figure 7, Case 7 shows how impact can occur since the use of a AND or exclusive OR gate indicates interactions between the different work-flow paths. Depending on the type of mission-specific interaction between the work-flow paths, impact may or may not occur.

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Depending on the mission thread's process work flow, where mission instances are located in the mission thread, and the timing of a cyber incident, one or more of the impact cases

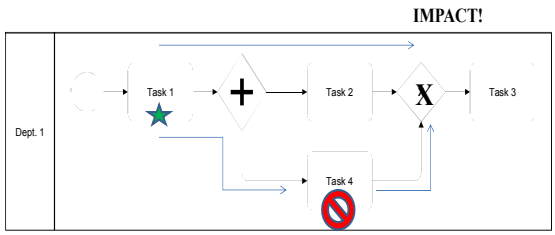

Figure 7: Case 7 – Impact may occur since an AND, or an exclusive OR join may require all or some of the paths to successfully complete

described above will occur. Any actual mission impact that occurs will be mission- and system-dependent, but the process modeling software allows these impact situations to be modeled in the following ways to produce the results presented in [2][3]:

1. Every ICT resource has attributes that reflect whether or not it may have been affected by a cyber incident.
2. A process model of the attack is run in parallel to the mission process to modify the affected resource(s) with the appropriate attack effect(s), starting at some time and persisting for some duration.

In a CMIA model, every ICT component, including data, is represented by a mission resource. Some of the resources can be allocated in a resource pool (e.g., a group of user workstations in a lab that are available on a first come, first serve basis), some individually, but all are assigned to unique ICT components.

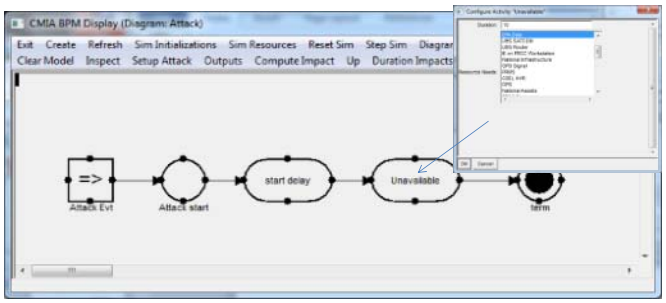

Figure 8: Process model representing an interruption (unavailability) of EPA data for 10 minutes

Interruption

For interruption (unavailability) attacks, such as a denial of service (DoS), a process model representing the attack mission thread runs in parallel with the mission process with an attack effect activity that grabs and then keeps the attack interrupted resource for the duration of the attack. When the incident is over, the attack thread releases the attack-interrupted resource, making it available again to other activities. While the attack mission

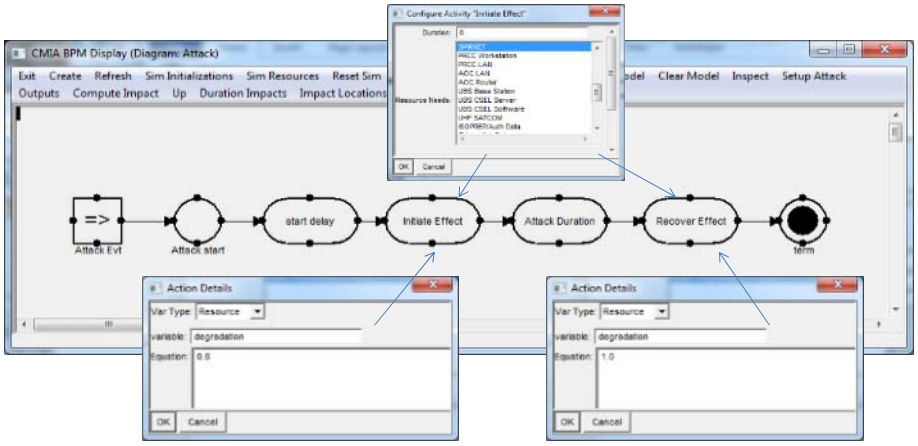

Figure 9: The degradation attribute of the attack-affected ICT resource is reduced to some fraction of its nominal value for the duration of an attack

3. Every ICT-dependent mission activity is annotated with its dependencies as to whether or not that activity could be successfully accomplished using an attack-affected resource. Those activity impacts are turned into mission-level impacts via the model.
4. The mission model is executed both with and without the cyber attack effects to compute MOE/MOPs and KPPs; mission impacts are reflected by changes in those performance parameters.

thread occupies the interrupted resource, it will not be available to any mission thread activities that might also need it.

When activities that are part of the MLM require the unavailable resource, they are unable to proceed until after the resource is released by the attack thread. The unavailability of a single resource will affect all mission work-flow paths that require use of the resource while the attack is in progress. The result of the unavailability of a resource will be delays in completing mission activities. The mission model reflects any mappings between the time it takes to perform tasks and mission outcomes. For example: When a person is lost at sea, the longer it takes to notify the coast guard (e.g., via some cyber-attack

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impacted communications channel), the less chance there will be to find that person alive. In e-commerce, delays in presenting products to a customer, or in processing their payment, can cause them to lose interest and to purchase their goods elsewhere. These mission effects are represented in the mission model to document the impacts of delays on mission outcomes.

Degradation

For degradation attacks, a process model representing the attack thread will set the degradation value of the attack affected resource to the value at the beginning of the attack and will reset it back to 1 when the attack is over. During the period of time that the attack mission thread has set the degradation attribute for the attack-affected resource to an impacted value other mission activities that use that resource may occur. If the resource represents a computer or a communications link, then the amount of simulation time taken to complete a mission activity that uses the resource will be increased in proportion to the amount of degradation. As with interruption attacks, other portions of the MLM model represent how temporal effects cause impacts as activities take longer to complete due to performance degradations.

If the attack-affected resource represents data, then logic or stochastic effects in the process model can represent how well mission activities can be performed given the level of degradation to the data resource. For example, an image analyst should be able to identify targets in a video stream from an unmanned aerial vehicle so long as a degraded image is not less than of 75% of its nominal quality, or an algorithm that converts speech to text will work so long as the audio signal is within 80% of nominal.

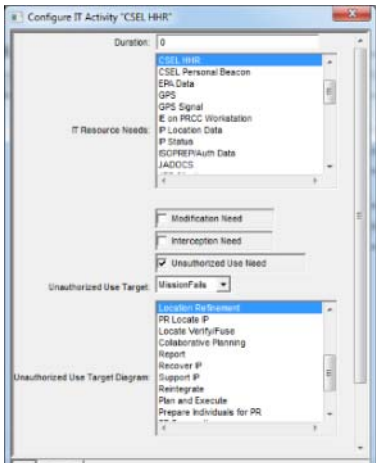

Figure 10: Built-in ability to configure process control flow given the cyber effects in IT activities

Interception, Modification, Fabrication, and Unauthorized Use Attacks

The remaining DIMFUI effects are modeled similarly to degradation, except that pre-defined attributes for interception, modification, fabrication, and unauthorized-use on the attacked resource are modified appropriately. The same attack model structure as is used for the degradation effect, except that "intercepted" is set to true for interception attacks, "modified" is set true for modification attacks, etc.

Whether or not a system's ICT resources cause mission impacts is system- and mission thread-specific. For each activity that relies on an ICT resource, the CMIA model captures whether or not the ability to perform the activity would be impacted if the ICT resource being used was suffering from a cyber-attack effect. Depending on the situation, some cyber-attack effects matter and some do not. For example, consider a GPS receiver being used in a mission system for timing and location information. When a GPS receiver is used for navigation, it's important that the GPS signal received by the receiver has not been modified. Similarly, it matters that the signal being received by the receiver is the actual GPS signal and is not a (spoofed) false one that could deceive the receiver about its location. It should also be apparent that although there are modification and fabrication constraints associated with the GPS signal, there are no mission constraints associated with the interception or unauthorized use of the GPS signal. Once the GPS signal is broadcast, anyone with a receiver is entitled to receive the GPS signal, and there are no restrictions on what they might do with it once they have received it. Our CMIA model would reflect that impacted outcomes occur with modification and fabrication of the GPS signal but that there are no impacts for its interception or unauthorized use.

In a related mission context, consider that the GPS receiver might be part of a hand-held radio that allows a downed pilot to communicate his location to a rescue center. In a civilian application, there is no need for the broadcast of the pilot location to be confidential. In a military use scenario, however, if enemy forces were able to intercept the location information being sent from the hand-held radio, that interception might be the difference between the capture or rescue of the pilot. Also, in the military use-case, an enemy's ability to capture and make an unauthorized use of the radio could allow it to send false information about the downed pilot, which could lead to bad outcomes for recovery forces. Explicitly representing these mission-specific constraints in the CMIA models provides a precise, objective description of the mission's cyber dependencies on cyber incident effects.

There are many possible examples of how mission outcomes can be impacted by cyber-attacks. There are also different ways these impacts can be modeled. On one hand, what should be captured in the model should reflect whether performing the mission activity while the ICT resource is affected by an attack would lead to an impact. For example, could an algorithm produce the correct result if its input data were to be modified? Would a mission to capture a terrorist succeed if the plan for the mission were leaked ahead of the mission?

Another way to consider how the effects of cyber incidents impact mission outcomes is to consider "implicit decisions" associated with detected incidents. In many circumstances there is an expectation that cyber incidents that have occurred will be detected. In these circumstances, the model should represent the implicit decisions that would happen at the mission level given those detections. By incorporating these "implicit" decisions, a CMIA model can be used for course of action (COA) analysis. For example, would an analyst complete her task less effectively if only three of five inputs were available, or would she abandon the task, or wait for all five to be available? By the same token, instead of waiting for a main server to become available, a

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backup server would be repurposed from its less critical task to act as a fallback replacement until the main server was up and running again.

IV. Custom Software

Previously we demonstrated how the techniques described above can be used to develop CMIA models that can predict mission impacts. We used the COTS software iGrafx to demonstrate the validity of the approach, but for a number of reasons, as we describe below, iGrafx lacked many features that we have automated in our software.

In CMIA models, every ICT resource includes the same set of DIMFUI attributes so that the mission impacts can be evaluated for mission activities that use those resources. In our software, resource attributes associated with the DIMFUI effects, along with their default values, are assigned automatically. We also provide default methods to allocate cyber resources from resource pools that ensure that resources not affected by cyber incidents are used first. By contrast, tools such as iGrafx require the attributes to be added manually for every IT resource, and implement only traditional resource allocation schemes such as FIFO and LIFO for resource pools.

Because we focus on ICT processes, we allow for the fact that the ICT resources can be imported from external sources such as network captures, scanners such as Nessus, or asset inventory products [12]. And since they are real objects, they can be assigned to organizations, owners, and locations. This allows them to be viewed from different perspectives, such as being organized topologically (the left diagram in Figure 2), making it easy to develop the ICT process the models from the diagrams. Searching on location, or organizational ownership constraints makes it manageable to find ICT resources when there are hundreds of resources from which to select. Products such as iGrafx do not support such views and filters, making it very hard to find the resources for which you are looking from an unordered list of hundreds of resource elements.

In iGrafx, the default mode of computation is performed via attributes in the process models. As a result, to convert from cyber effects into mission effects, we developed a structured condition handler process model to use as a connector to bridge between the ICT process models and the MLM model that represents the mission activities. Since there was no way to automatically create these connector models in iGrafx, we made an original, which we had to copy repeatedly every time we added a new ICT process model to a MLM. Linkage between the ICT process model diagrams, the connectors, and the MLM diagrams had to be organized and managed via a naming convention. In iGrafx all of this can work, but it requires the discipline of following a specific convention by the modelers to keep everything straight. This makes understanding someone else's model, or a model that was developed months earlier, much more challenging than it should be. To address these challenges, in our software each ICT activity is preprogrammed with logic that eliminates the need for connectors and comes with a custom GUI (Figure 10) that supports catch and throw activities (i.e., activities not linked by wires) attached to DIMFUI effects. These allow the control flow of the model to explicitly pass to parts of the model that deal with the converting of ICT DIMFUI effects into mission effects, or that represent implicit decision or failover processes. The result is CMIA mission models that are cleaner to look at and not cluttered with extra process logic. This makes them easier to develop, easier to understand, and easier to maintain. Also, since some modeling knowledge is procedural in nature, the CMIA software is extensible via a scripting language so that models or the system can be extended as needed.

Since the purpose of our software is to be able to compute the impacts of different cyber incidents, it has dialogs and functions specific to that purpose. A GUI allows a user to specify the details of a cyber incident. Each incident involves a DIMFUI effect against an ICT resource, a starting time, and a duration. Multiple attack effects can be entered if necessary, and if the incident duration is not known a different GUI enables impacts to be estimated for a set of hypothetical durations. Behind the scenes, using the predefined structure of cyber incident models (described in the previous section), attack models are automatically created. After that a button click runs the model, both with and without the cyber incident effects, and presents a dialog that shows the impact results. Since some MLM models can contain stochastic effects, computing impacts with them can require multiple trials to compute; our software takes care of this automatically. In contrast, when using iGrafx, the attack models themselves dealt with the processing of multiple runs, and a custom GUI had to be manually developed for each and every mission model.

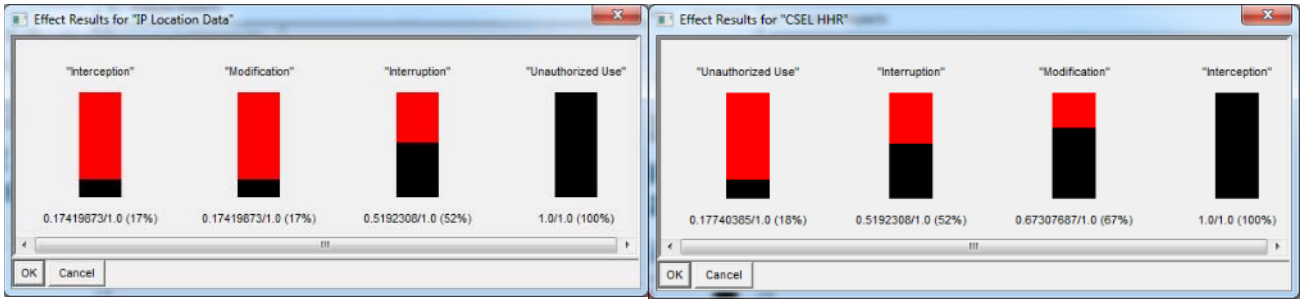

Figure 11: Given any selected IT resource assessment, results show which incident effects cause mission impacts

As documented in [3], these CMIA models can be used for analyzing a mission system's susceptibility to cyber incidents. This analysis also serves as a way to help validate the models. It involves running every attack effect against every ICT resource, with different durations and timings. In iGrafx this is possible

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using their Six Sigma product and a MINITAB [9] add-on, but once again that requires adding a lot of extra process logic to the model to support the computing set of cyber incidents. In our CMIA software these functions are already built-in. Also included is the ability to view the impact results and inspect the results by effect, resource, etc. (Figure 11). In iGrafx, the results had to be exported to a spreadsheet to be analyzed.

Our CMIA software also has model inspection and debugging capabilities, allowing a developer or user to step through execution of the model, inspect attributes and confirm its operations.

V. DISCUSSION

In previous papers we discussed how the characteristics of mission impacts motivated our choosing a process modeling approach [1]. Our experience to date is that while building the process models of mission and ICT manually can be tedious, it is not difficult. What is difficult is obtaining the necessary information about the mission and its ICT dependencies. Since impacts can be task-specific, system capability-oriented, immediate, or predicted as a future event, our process modeling approach is the only currently practiced technique that supports all of these assessments. So, for example, "playing forward" from an incident can predict cascading impacts that may occur. Techniques in competition with CMIA, such as dependency tree-based mission representations and those using rollup rules [11], are not as capable [1] since they support a much more limited view of what impact is.

The original intended purpose for the CMIA methodology was to provide a decision aid that would help mission personnel understand the mission implications of observed cyber incidents. This use case depends on the ability to monitor ones network, detect the incidents that occur, and generate incident reports that describe which cyber effects affect which cyber components. When possible, it helps to have an estimate for the duration of the event or a time-to-recovery. An example of this type of reporting process is described in [10]. Our software can also be used as a cyber mission course of action (COA) tool to help mission personnel assess whether there are fallback or failover processes that are better alternatives than waiting for the incident to end or be recovered from.

CMIA models must be able to assess any possible incident that can occur, so they focus on the impacts that the incident causes rather than whether or not the incident is likely to occur or not (i.e., whether the system is vulnerable). This makes sense since, given the steady development of 0-day exploits, even a fully patched system can still be vulnerable. This focus on a mission outcomes' susceptibility to cyber effects differentiates our approach from others that focus primarily on known vulnerabilities. Our CMIA approach and tool is mission-centric rather than attacker- or vulnerability-centric.

A primary reason for building our tool is to make it much easier to build, evaluate, and maintain CMIA mission to cyber correspondence models. As described in [1], these types of models are fundamental to realizing many of the mission assurance capabilities, and are useful across a system's entire lifecycle. Other approaches to mission assurance analysis (e.g., a crown jewels analysis) often leave too many of the details associated with the analysis (e.g., failover and workarounds) only in the head of the analyst, or as separate items only adjunct to the method. As a result, when the system or mission characteristics change it is not trivial to update them, especially if the original modeler is no longer around. What is needed is a method that captures these mission and cyber details as a complete, precise, objective description of the mission system and that documents the interactions among mission resources and events that affect outcomes. Our CMIA approach accomplishes this by leveraging capabilities developed and validated by the business process modeling community to describe missions. Moreover, it extends them to include ICT processes. Our CMIA software implements the approach by codifying and automatically including many of the modeling and support functions that would otherwise have to be added manually. The result is that the models are simpler and easier to build and maintain than they would be if they were built with general purpose BPM software. Currently the CMIA tool can compute the mission impacts of cyber incidents, both real and hypothetical [2], and can be used to perform a crown jewels analysis [3]. Forthcoming work will show how it can be combined with a topological attack model to perform additional cyber risk assessments.

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